University of Europe for Applied Sciences

Application for Capstone Project (W/2026-27)

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Study Program

M.Sc. Data Science

Title of Project

VeriCBAM: Multimodal Evidence Fusion & Decision Support for CBAM Emissions Consistency Assessment

Short Description (2-3 Paragraphs)

In 2026, the EU Carbon Border Adjustment Mechanism (CBAM, Regulation (EU) 2023/956) entered its definitive financial enforcement phase, obliging European importers of carbon-intensive commodities (cement, iron and steel) to declare embedded direct emissions from non-EU installations under Commission Implementing Regulation (EU) 2025/2547, subject to third-party verification under Commission Implementing Regulation (EU) 2025/2546 and Commission Delegated Regulation (EU) 2025/2551. Importers and authorities face an acute screening bottleneck: supplier declarations require substantial audit scrutiny, while physical on-site audits are economically and logistically constrained (~12,000 import entities competing for limited accredited verifiers).

This project develops VeriCBAM, an explainable multimodal decision-support system that screens and assesses the consistency of CBAM declarations against independent physical, observational, and historical evidence. VeriCBAM integrates three empirical pillars: (1) a first-principles stoichiometric and engineering engine establishing process-specific lower bounds for cement clinker calcination and metallurgical steelmaking (BF-BOF vs. Scrap-EAF); (2) a Copernicus Earth Observation pipeline extracting real Sentinel-5P TROPOMI Level-2 tropospheric NO2 column enhancements as an operational activity indicator (with CO and Sentinel-2 SWIR planned as extensions); and (3) longitudinal historical facility fingerprints derived from independent emissions registries (EEA E-PRTR).

The core decision layer features an expert-parameterized probabilistic evidence-fusion engine that quantifies material inconsistency risk alongside separate

- [x] I am aware of the workload requirements for this module
- [ ] I will conduct this project together with an outside partner
- [ ] I am opening the project for potential teamwork
- [ ] The project is under an NDA

Place, Date

Berlin, 12.10.2026

Signature

For internal use

- [ ] Project approved
- [ ] Project proposal requires revision

Place, Date

Signature